

James J. Augsburger
Susan Schneider
Jorge Freire
Luther W. Brady

Survival following enucleation versus plaque radiotherapy in statistically matched subgroups of patients with choroidal melanomas: results in patients treated between 1980 and 1987

Received: 1 September 1998
Revised version received: 5 October 1998
Accepted: 26 October 1998

This work was presented in part at the XXIst Meeting of the Club Jules Gonin, Edinburgh, Scotland, 28 August–1 September 1998

J. J. Augsburger¹ (✉) · S. Schneider
Oncology Unit, Retina Service,
Wills Eye Hospital,
Jefferson Medical College,
Thomas Jefferson University,
900 Walnut Street, Philadelphia,
PA 19107-5598, USA
Tel. +1-215-928-3221
Fax +1-215-928-3122

J. Freire · L. W. Brady
Department of Radiation Oncology
and Nuclear Medicine,
Allegheny University of the Health Sciences,
Hahnemann Division, Philadelphia,
Pennsylvania, USA

Present address:

¹ Department of Ophthalmology,
University of Cincinnati College of Medicine,
Health Professions Building, Suite 350,
Eden & Bethesda Avenues,
Cincinnati, OH 45267-0527, USA

Abstract ● **Background:** The relative effectiveness of plaque radiotherapy and enucleation in patients with choroidal or ciliochoroidal melanoma remains controversial. Most published comparative survival studies have evaluated clearly biased groups, short-term follow-up, or both. ● **Methods:** The authors performed a retrospective, non-randomized, comparative survival study on 734 patients with a choroidal or ciliochoroidal melanoma treated by enucleation ($n=237$) or plaque radiotherapy ($n=497$) between July 1980 and June 1985, and on a residual group of 448 patients, 103 treated by enucleation and 345 treated by plaque radiotherapy, identified by variable-by-variable range matching. Melanoma-specific mortality and all-cause mortality were evaluated by the life-table method. ● **Results:** The melanoma-specific mortality rate was substantially worse in the original enucleation subgroup throughout available

post-treatment follow-up; however, differences in baseline prognostic factors between the subgroups are likely to explain much of the survival differential. After eliminating patients with nonoverlapping values of individual clinical variables to adjust for recognized intergroup differences at baseline, there was no significant difference in the mortality curves of the residual subgroups. The cumulative 15-year survival based on deaths from metastatic melanoma in the enucleation subgroup was 57.1% (standard error=6.4%), while that in the plaque radiotherapy subgroup was 61.8% (standard error=3.3%).

● **Conclusion:** On the basis of these results, we conclude that a large difference in survival between equivalent groups of patients with primary choroidal or ciliary body melanoma treated by enucleation or by plaque radiotherapy appears unlikely.

Introduction

Over 12 years have passed since the Collaborative Ocular Melanoma Study began to enroll patients in its plaque radiotherapy versus enucleation trial for medium-size choroidal melanomas [15]. During this time, numerous peer-reviewed articles, book chapters and editorials on plaque radiotherapy versus enucleation for choroidal and ciliochoroidal melanomas have been published. In spite of these publications, the relative effec-

tiveness of plaque radiotherapy and enucleation remains controversial. Although evidence derived from a number of nonrandomized, retrospective, comparative survival studies that have incorporated statistical adjustment for recognized baseline susceptibility bias suggests that a large difference in melanoma-specific mortality rates between the treatment groups is unlikely [1–5, 8], there is still concern that patients treated by plaque radiotherapy may suffer a delayed downturn in survival related to local tumor relapses, known to occur in some patients [9, 16].

We recently reported 15-year survival results on a previously reported group of 237 patients with choroidal or ciliary body melanoma treated by enucleation ($n=140$) or cobalt-60 plaque radiotherapy ($n=97$) [5]. All of these patients had been treated between the interval May 1976 and June 1980. Our analysis found no late downturn in the survival curve of the plaque-treated patients and no clinically important difference between the all-cause mortality or melanoma-specific mortality curves of the patients in the treatment subgroups. Because the results of this recently published follow-up study are surprising to some proponents of enucleation, we decided to evaluate the comparative survival of independent subgroups of patients with choroidal or ciliochoroidal melanoma treated by plaque radiotherapy or by enucleation. Our objective was to confirm or contradict the results of our recently published study. We evaluated the long-term survival distributions of a previously unreported group of patients with primary choroidal or ciliary body melanoma treated by plaque radiotherapy or enucleation in our center between 1980 and 1985.

Patients and methods

Patients

The patients evaluated in this study consisted of a previously unreported group of 734 persons with a clinically diagnosed primary choroidal or ciliochoroidal melanoma treated by either enucleation ($n=237$) or plaque radiotherapy ($n=497$) at one center between 1 July 1980, and 30 June 1985. Several different isotopes were used in plaque radiotherapy in our center during the study period [7]. In this group, the isotope used was cobalt-60 in 382, iridium-192 in 80, indium-125 in 18, and ruthenium-106 in 17 patients. For the purposes of this study, the several types of plaque were assumed to be equally effective.

Clinical variables

Multiple clinical variables were assessed prospectively for each patient. The variables evaluated and the methods of their assessment have been described [2–5]. The units of dimensional variables and categories of nominal and ordinal variables were the same as those reported in our recently published paper on long-term survival following enucleation versus cobalt-60 plaque radiotherapy [5].

Treatment assignment

Treatment by plaque radiotherapy or enucleation was not randomized but assigned selectively on a case-by-case basis. Factors influencing the treatment assignment included tumor size, tumor location, presence of extrascleral tumor extension, visual status of the affected eye, visual status of the fellow eye, extent of retinal detachment, age and general health of the patient, and patient and physician preference for a particular treatment. For our survival data analysis, treatment by enucleation was coded 0 and treatment by plaque was coded 1.

Patient follow-up

The survival status of each patient and the duration of each patient's post-treatment follow-up until death or most recent evaluation were determined by retrospective review of the clinical records of those who continued to be followed in our center, by mail or telephone communication with those no longer being followed in our center (or with members of their families), by mail or telephone communication with local physicians involved in the care of the patients, and by analysis of national mortality data from the National Death Index of the United States. For patients who had died, we attempted to determine the cause of death by communicating with the physicians caring for them at the time of death and by obtaining and reviewing death certificates. Survival status of each patient at most recent follow-up was categorized as dead of metastatic uveal melanoma, dead of nonmelanoma cancer, dead of noncancer cause, or alive. Patients dying with metastatic uveal melanoma were considered to have died of their metastatic disease, even if that metastatic disease was determined not to be the immediate cause of death. The date of survey for this study was 1 February 1998.

Comparison of survival in study subgroups

Actuarial survival curves were computed for the original subgroups of patients treated by enucleation and plaque radiotherapy using the life-table method [10]. Endpoint events evaluated in this survival analysis were (1) death from metastatic uveal melanoma and (2) death from any cause. The survival curves of the subgroups were compared using the Mantel-Haenszel test [11].

Assessment of similarity of patient subgroups

The treatment subgroups were compared with respect to the dichotomous or multichotomous distributions of all study variables using Pearson's chi-squared test. In addition, the distributions of all normally distributed scalar variables in the subgroups were compared using the two-tailed unpaired *t*-test. In our prior publication [5], we noted that there were substantial differences between the treatment subgroups in terms of many of the assessed clinical variables, including tumor size (larger in the enucleation subgroup), location of the anterior tumor margin relative to the equator and ora serrata (higher proportion of tumors involving ciliary body in the enucleation subgroup), and patient age (younger patients in the enucleation subgroup). Because these variables were also factors that influenced our treatment recommendation, we anticipated finding a similar susceptibility bias between the treatment subgroups in favor of the plaque radiotherapy subgroup in this study.

Elimination of patients to form residual subgroups

To reduce the magnitude of baseline susceptibility bias between the subgroups, patients in either subgroup who had values of individual clinical variables that were substantially greater than or less than those of any patients in the other subgroup were eliminated from the analysis group. The precise method used to identify nonoverlapping values of individual variables and eliminate patients was as follows: (1) frequency distributions of all study variables were printed for each subgroup; (2) the printouts were compared to identify the lowest and highest values of each variable in the two subgroups; (3) when minimal or maximal values of an individual variable were found to be substantially different in the subgroups, arbitrary cutoff values for that variable were selected such that both subgroups had patients with values of the variable just above the lower cutoff value and just below the upper cutoff value; and (4) patients having one or more variables with values outside the selected cutoff limits were excluded to form residual subgroups.

Assessment of similarity of residual subgroups

The residual subgroups were compared with respect to the dichotomous or multichotomous distributions of all study variables using Pearson's chi-squared test. In addition, the distributions of all normally distributed scalar variables in the residual subgroups were compared using the two-tailed unpaired *t*-test.

Comparison of actuarial survival curves in residual subgroups

Actuarial survival curves based on deaths from metastatic uveal melanoma were computed for the residual subgroups using the life-table method [10]. These subgroup survival curves were compared using the Mantel-Haenszel test [11].

Estimating adjusted relative impact of plaque radiotherapy and enucleation

The adjusted relative impact of plaque radiotherapy versus enucleation was estimated for the combined group of residual patients by forcing the treatment variable (enucleation coded 0, plaque radiotherapy coded 1) into a comprehensive multivariate Cox proportional hazards model that incorporated all evaluated study variables (except that it included only one of the several tumor size variables, largest linear tumor dimension), determining the coefficient and standard error of the coefficient for the treatment variable in that model, and computing the rate ratio and 95% confidence interval for the treatment variable from those values [12].

Significance level and power calculations

A nominal alpha level of 0.05 was specified prior to data analysis as the cut-off value for assessing the significance of differences between subgroups in our comparative statistical tests. Because of the multiplicity of statistical contrasts in this study, we actually interpreted only those contrasts associated with a *P* value less than 0.01 as significant. The power of our comparative survival study to show various differences in survival curves between the residual subgroups was estimated by the method of Feinstein [6]. These computations were performed using the QC Tools routine of the MEDLOG Database Management and Analysis System (MEDLOG Systems, Information Analysis, Incline Village, Nev.).

Results

Summary of survival status in original patient subgroups

Table 1 shows the last known survival status of all of the patients in the two original patient subgroups. The median follow-up time for surviving patients in the original group of 734 patients was 13.9 years. Only 14 surviving patients in the plaque-treated subgroup (2.8%) and 8 surviving patients in the enucleation subgroup (3.4%) had been followed up for less than 10 years.

Survival distributions of original patient subgroups

Figure 1 shows the life-table survival curves of the two subgroups over 15 years of post-treatment follow-up. There appears to be a substantial separation between the curves based on deaths from metastatic melanoma, and this difference approaches statistical significance (Mantel-Haenszel chi-squared=3.5, *P*=0.062). In contrast, the survival curves based on deaths from any cause are more nearly identical (Mantel-Haenszel chi-squared=0.5, *P*=0.48). Note that there is no precipitous downturn in the survival curve of the plaque subgroup after the 10th post-treatment year.

Dissimilarity of patients in original subgroups

Table 2 summarizes the differences between the original subgroups in terms of the evaluated patient and tumor variables. Inspection of this table shows that the original subgroups were substantially dissimilar at baseline with respect to many of the evaluated variables. Specifically, patients in the enucleation subgroup were substantially younger and tended to have more symptoms, worse pre-treatment visual acuity, more extensive retinal detachment, and larger and more anteriorly located tumors than did patients selected for plaque therapy. The susceptibility bias revealed by this analysis indicates that direct compar-

Table 1 Last known survival status of all patients in the original treatment subgroups

Survival status	Patient subgroup	
	Plaque radiotherapy (<i>n</i> =497) No. (%)	Enucleation (<i>n</i> =237) No. (%)
Alive		
Without metastatic uveal melanoma	280 (56.4)	130 (54.8)
With metastatic uveal melanoma	9 (1.8)	2 (0.8)
Dead		
Dead of metastatic uveal melanoma	165 (33.2)	90 (38.0)
Dead of nonmelanoma cancer	11 (2.2)	7 (3.0)
Dead of noncancer cause	32 (6.4)	8 (3.4)

Table 2 Distributions of evaluated baseline clinical variables by treatment subgroups in original group of 734 patients with primary choroidal or ciliary body melanoma

Variable	Treatment subgroup				df for chi-squared test	Chi-squared ^b or t value ^c	P
	Plaque Radiotherapy (n=497)		Enucleation (n=237)				
	No. (%)	Mean ^a	No. (%)	Mean ^a			
Age (years)							
≤40	42 (8.5)		60 (25.3)				
>40 to ≤50	70 (14.1)		25 (10.5)				
>50 to ≤60	134 (27.0)		55 (23.2)		4	39.6 ^b	<0.001
>60 to ≤70	138 (27.8)		60 (25.3)				
>70	113 (22.7)		37 (15.6)				
Mean		59.2		54.5		4.3 ^c	<0.001
Sex							
Male	247 (49.7)		117 (49.4)				
Female	250 (50.3)		120 (50.6)		1	0.0 ^b	0.99
Symptoms							
Absent	91 (18.3)		12 (5.1)				
Present	406 (81.7)		225 (94.9)		1	24.7 ^b	<0.001
Visual acuity							
20/40 or better	320 (64.4)		76 (32.1)				
20/50 to 20/200	127 (25.6)		66 (27.8)		2	106.1 ^b	<0.001
worse than 20/200	50 (10.0)		95 (40.1)				
Intraocular pressure (mmHg)							
<12	108 (21.7)		56 (23.6)				
≥12 to <18	299 (60.2)		129 (54.4)		2	1.7 ^b	0.43
≥18	90 (18.1)		52 (21.9)				
Mean		14.3		16.6		4.1 ^c	<0.001
Anteroposterior basal tumor diameter (mm)							
≤10	173 (34.8)		56 (23.6)				
>10 to ≤15	250 (50.3)		110 (46.4)		2	28.1 ^b	<0.001
>15	74 (14.9)		71 (30.0)				
Mean		11.1		12.5		5.1 ^c	<0.001
Circumferential basal tumor diameter (mm)							
≤10	241 (48.5)		80 (33.8)				
>10 to ≤15	213 (42.9)		109 (46.0)		2	25.7 ^b	<0.001
≥15	43 (8.6)		48 (20.2)				
Mean		10.0		11.4		5.0 ^c	<0.001
Largest basal tumor diameter (mm)							
≤10	158 (31.8)		53 (22.4)				
>10 to ≤15	256 (51.5)		106 (44.7)		2	29.6 ^b	<0.001
>15	83 (16.7)		78 (32.9)				
Mean		11.4		12.7		5.0 ^c	<0.001
Thickness (mm)							
≤5	189 (38.0)		52 (21.9)				
>5 to ≤8	187 (37.6)		79 (33.3)		2	37.0 ^b	<0.001
>8	121 (24.4)		106 (44.8)				
Mean		6.2		7.6		6.3 ^c	<0.001
Largest linear tumor dimension (mm)							
≤10	158 (31.8)		52 (21.9)				
>10 to ≤15	255 (51.3)		107 (45.1)		2	28.9 ^b	<0.001
>15	84 (16.9)		78 (33.0)				
Mean		11.4		12.7		5.0 ^c	<0.001
Location of anterior tumor margin							
Posterior to equator	200 (40.2)		108 (45.6)				
Between equator and ora	155 (31.2)		57 (24.0)		2	9.0 ^b	0.011
Anterior to ora serrata	142 (28.6)		72 (30.4)				

Table 2 (continued)

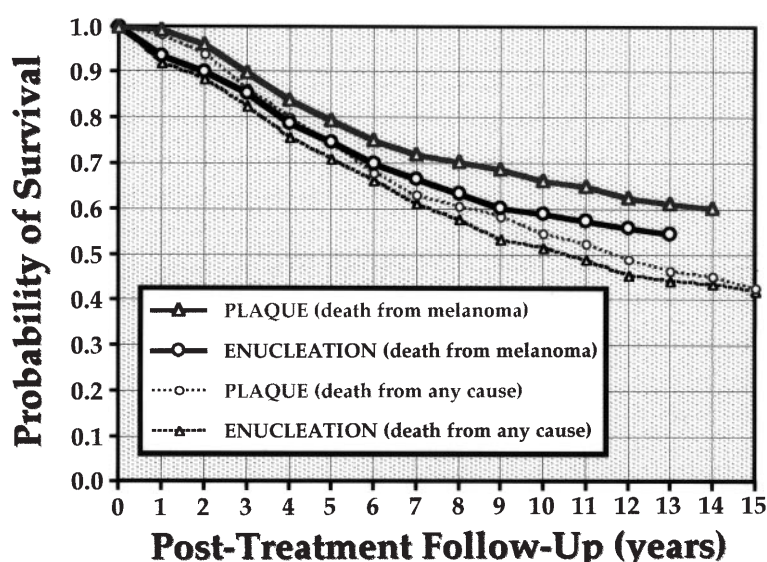
Variable	Treatment subgroup		df for chi-squared test	Chi-squared ^b or t value ^c	P
	Plaque Radiotherapy (n=497)	Enucleation (n=237)			
	No. (%)	Mean ^a	No. (%)	Mean ^a	
Location of posterior tumor margin relative to optic disc					
At disc margin	31 (6.2)		77 (32.5)		
≤3 mm from disc	128 (25.8)		70 (29.5)		
>3 mm from disc	338 (68.0)		90 (38.0)		
			2	101.5 ^b	<0.001
Location of posterior tumor margin relative to foveola					
Involving foveola	31 (6.2)		55 (23.2)		
≤3 mm from foveola	151 (30.4)		83 (35.0)		
>3 mm from foveola	315 (63.4)		99 (41.8)		
			2	53.4 ^b	<0.001
Extent of retina detached (%)					
None	127 (25.6)		27 (11.4)		
>0 to ≤25	214 (43.0)		83 (35.0)		
>25 to ≤50	88 (17.7)		53 (22.4)		
>50	68 (13.7)		74 (31.2)		
			3	48.1 ^b	<0.001
Mean		21.9		39.5	8.1 ^c
					<0.001

^a Means are shown only for scalar variables recorded on a continuous numeric scale

^b Chi-squared values pertain to comparisons of discontinuous distributions of variables

^c Unpaired t-values pertain to comparisons of means of continuous scalar variables

Fig. 1 Life-table survival curves based on melanoma-specific deaths (*solid lines*) and all causes of death (*dashed lines*) in original patient subgroups. Curves were plotted through the last interval during which at least one event occurred



ison of the unadjusted survival curves of these two subgroups, as shown in Fig. 1, is invalid for determining the relative effectiveness of the two therapies.

Patients excluded for nonoverlapping values of individual variables

Table 3 shows the cut-off values (minimum and maximum) for every variable used to exclude patients, the

number of patients excluded from each original subgroup because of nonoverlapping values of the specified variables, and the total number of patients excluded from each subgroup. These exclusions left us with residual subgroups of 235 patients treated by plaque and 103 patients treated by enucleation.

Table 3 Number of patients excluded from two original subgroups by sequential range matching of individual clinical variables. Variables are listed in the order in which the range-matching comparisons were performed. Some patients had more than one reason for exclusion; however, only the first encountered reason for exclusion is presented here. Study variables not listed in this table had completely overlapping ranges of values and were not used to exclude any patients

Variable Values that prompted exclusion	Original plaque radiotherapy subgroup (<i>n</i> =497) Number excluded	Original enucleation subgroup (<i>n</i> =237) Number excluded
Clinical extraocular extension		
Any extraocular extension	3	16
Prior treatment of intraocular tumor		
Any prior treatment	5	2
Prior intraocular surgery on affected eye		
Any intraocular surgery within 6 months	11	4
Metastatic uveal melanoma status at baseline		
Presence of metastatic disease	2	0
Laterality of uveal melanoma		
Bilateral uveal melanoma	1	0
Symptoms attributable to tumor		
Blind painful eye	4	7
Visual acuity in affected eye		
≤hand motion	7	6
Clarity of optical media		
Opaque media	0	1
Location of posterior tumor margin		
Posterior tumor margin anterior to ora serrata	2	1
Location of anterior tumor margin		
Anterior tumor margin involving iris	12	9
Diameter of tumor in antero-posterior direction		
<7 mm or >16 mm	48	36
Diameter of tumor in circumferential direction		
<5 mm or >15 mm	13	9
Thickness of tumor		
<2.5 mm or >12 mm	14	5
Age of patient at time of treatment		
<30 years or >80 years	14	8
Intraocular pressure		
<8 mm Hg or >24 mm Hg	10	4
Retinal detachment		
>80% detached	6	3
Prognostic index (calculated from Cox proportional hazards model)		
<0.15 or >5.5	0	2
Total number of patients excluded	152	134

Comparison of residual patient subgroups

Comparison of the residual patient subgroups (Table 4) reveals considerably greater similarity between these subgroups than between the original patient subgroups. In spite of this, potentially important differences in several prognostic baseline clinical variables still exist. Patients in the enucleation subgroup were still somewhat younger and still tended to have more symptoms, worse pretreatment visual acuity, more extensive retinal detachment, and larger and more anteriorly located tumors than did patients selected for plaque therapy. The differences revealed by this analysis imply that our attempts to reduce susceptibility bias between the subgroups were not completely effective.

Summary of survival status in residual patient subgroups

Table 5 summarizes the last known survival status of the 448 patients in the residual group. The median follow-up time for surviving patients in the residual plaque subgroup was 14.2 years, and that for the surviving patients in the enucleation subgroup was 13.3 years. Only 10 surviving patients in the residual plaque subgroup (2.9%) and 4 surviving patients in the residual enucleation subgroup (3.9%) had been followed up for less than 10 years.

Survival distributions of residual patient subgroups

Figure 2 shows the life-table survival curves of the two residual subgroups based on deaths from metastatic melanoma and all causes. The survival curve based on deaths from metastatic melanoma for the patients in the plaque

Table 4 Distributions of baseline clinical variables by treatment subgroups in residual group of 448 patients with primary posterior uveal melanoma

Variable	Treatment subgroup				df for chi-squared test	Chi-squared ^b or t value ^c	P
	Plaque radiotherapy (n=345)		Enucleation (n=103)				
	No. (%)	Mean ^a	No. (%)	Mean ^a			
Age (years)							
≤40	27 (7.8)		27 (26.2)		4	28.2 ^b	<0.0001
>40 to ≤50	53 (15.4)		15 (14.6)				
>50 to ≤60	101 (29.3)		23 (22.3)				
>60 to ≤70	96 (27.8)		28 (27.2)				
>70	68 (19.7)		10 (9.7)				
Mean		58.5		53.0		4.1 ^c	<0.001
Sex							
Male	184 (53.3)		52 (50.5)		1	0.16 ^b	0.69
Female	161 (46.7)		51 (49.5)				
Symptoms							
Absent	64 (18.6)		3 (2.9)		1	14.0 ^b	0.0002
Present	281 (81.4)		100 (97.1)				
Visual acuity							
20/40 or better	245 (71.0)		40 (38.8)		2	40.7 ^b	<0.0001
20/50 to 20/200	79 (22.9)		41 (39.8)				
Worse than 20/200	21 (6.1)		22 (21.4)				
Intraocular pressure (mm Hg)							
<12	66 (19.1)		28 (27.2)		2	4.3 ^b	0.12
≥12 to <18	218 (63.2)		63 (61.2)				
≥18	61 (17.7)		12 (11.7)				
Mean		14.3		13.6		2.0 ^c	0.046
Anteroposterior basal tumor diameter (mm)							
≤10	113 (32.8)		27 (26.2)		2	3.9 ^b	0.15
>10 to ≤15	197 (57.1)		59 (57.3)				
>15	35 (10.1)		17 (16.5)				
Mean		11.0		11.5		1.7 ^c	0.087
Circumferential basal tumor diameter (mm)							
≤10	176 (51.0)		46 (44.7)		2	3.3 ^b	0.20
>10 to ≤15	160 (46.4)		51 (49.5)				
>15	9 (2.6)		6 (5.8)				
Mean		9.7		10.2		1.7 ^c	0.093
Largest basal tumor diameter (mm)							
≤10	104 (30.1)		26 (25.2)		2	3.6 ^b	0.16
>10 to ≤15	204 (59.1)		59 (57.3)				
>15	37 (10.7)		18 (17.5)				
Mean		11.2		11.6		1.4 ^c	0.15
Thickness (mm)							
≤5	131 (38.0)		23 (22.3)		2	9.9 ^b	0.0072
>5 to ≤8	140 (40.6)		47 (45.6)				
>8	74 (21.4)		33 (32.0)				
Mean		6.0		6.7		2.5 ^c	0.012
Largest linear tumor dimension (mm)							
≤10	104 (30.1)		25 (24.3)		2	3.9 ^b	0.14
>10 to ≤15	204 (59.1)		60 (58.3)				
>15	37 (10.7)		18 (17.5)				
Mean		11.2		11.6		1.6 ^c	0.12
Location of anterior tumor margin							
Posterior to equator	145 (42.0)		59 (57.3)		2	7.4 ^b	0.024
Between equator and ora	117 (33.9)		26 (25.2)				
Anterior to ora serrata	83 (24.1)		18 (17.5)				

Table 4 (continued)

Variable	Treatment subgroup				df for chi-squared test	Chi-squared ^b or t value ^c	P
	Plaque radiotherapy (n=345)		Enucleation (n=103)				
	No. (%)	Mean ^a	No. (%)	Mean ^a			
Location of posterior tumor margin relative to optic disc							
At optic disc margin	17 (4.9)		34 (33.0)		1	68.2 ^b	<0.0001
≤3 mm from disc	92 (26.7)		31 (30.1)				
>3 mm from disc	236 (68.4)		38 (36.9)				
Location of posterior tumor margin relative to foveola							
Beneath foveola	14 (4.1)		21 (20.4)		1	32.5 ^b	<0.0001
≤3 mm from foveola	109 (31.6)		36 (35.0)				
>3 mm from foveola	222 (64.3)		46 (44.7)				
Extent of retina detached (%)							
None	75 (21.7)		10 (9.7)		3	19.2 ^b	0.0002
>0 to ≤25	154 (44.6)		35 (34.0)				
>25 to ≤50	74 (21.4)		34 (33.0)				
>50	42 (12.2)		24 (23.3)				
Mean		22.0		34.6		4.9 ^c	<0.001

^a Means are shown only for scalar variables recorded on a continuous numeric scale

^b Chi-squared values pertain to comparisons of discontinuous distributions of variables

^c Unpaired t-values pertain to comparisons of means of continuous scalar variables

Table 5 Last known survival status of all patients in the residual treatment subgroups

Survival status	Patient subgroup	
	Plaque radiotherapy (n=345) No. (%)	Enucleation (n=103) No. (%)
Alive		
Without metastatic uveal melanoma	200 (58.0)	62 (60.3)
With metastatic uveal melanoma	9 (2.6)	2 (1.9)
Dead		
Dead of metastatic uveal melanoma	105 (30.4)	34 (33.0)
Dead of nonmelanoma cancer	8 (2.3)	3 (2.9)
Dead of noncancer cause	23 (6.7)	2 (1.9)

subgroup is still higher than that of the patients in the enucleation subgroup throughout the entire follow-up; however, the magnitude of the separation between the curves is considerably less than that shown in Fig. 1. The difference between these curves is not statistically significant (Mantel-Haenszel chi-squared=0.52, $P=0.47$).

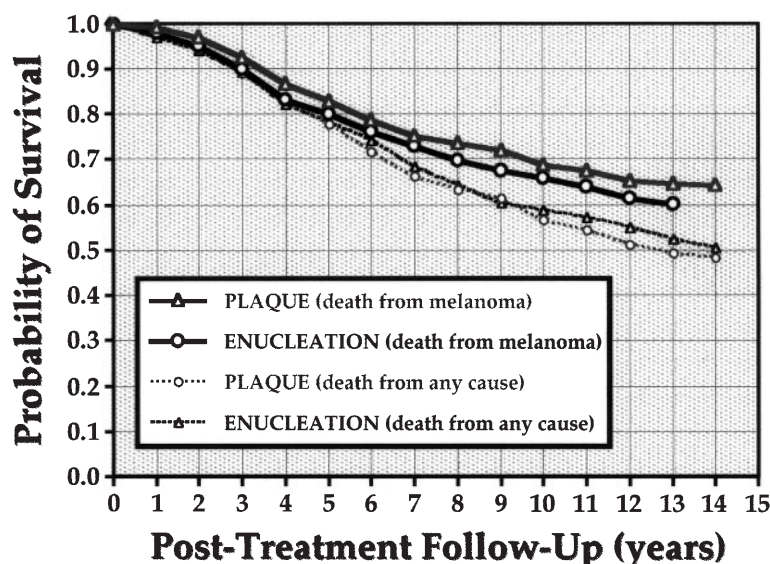
In the residual plaque subgroup, 43 of the 345 patients (12.5%) developed local tumor relapse during available follow-up. The 15-year life-table event rate for local relapse was 18.9%. The 15-year life-table survival based on deaths from all causes was 43.7% in the patients who experienced local relapse but 65.1% in the patients who exhibited satisfactory sustained local tumor regression (Mantel-Haenszel chi-squared=9.0, $P=0.0027$). In spite of this survival differential, note that there is no delayed downturn in the survival curve of the residual plaque

subgroup shown in Fig. 2 or late crossing of the curves.

Adjusted relative rate ratio for treatment variable

Adding the treatment variable to a comprehensive multivariate Cox proportional hazards model applied to our combined residual patient group yielded a coefficient for the treatment variable of -0.14 (standard deviation=0.17). The relative rate ratio computed from the coefficient was 0.87. This value, less than 1.0, indicates that the adjusted relative likelihood of dying from uveal melanoma metastasis was slightly lower in the plaque radiotherapy subgroup. However, the 95% confidence interval of the relative rate ratio, which extends from

Fig. 2 Life-table survival curves based on melanoma-specific deaths (*solid lines*) and all causes of death (*dashed lines*) in residual patient subgroups. Curves were plotted through the last interval during which at least one event occurred



0.62 to 1.22, shows that this difference is not statistically significant.

Power of study

Given the relatively small number of cases in the residual treatment subgroups, it should not be surprising that the power of this study to show a difference as small as that actually observed is low (less than 50%). However, the power of our study to show a difference of 50% or more between the subgroups is approximately 80%. This result suggests that a large treatment differential between plaque radiotherapy and enucleation is unlikely.

Discussion

The results of this study indicate that (1) there is no precipitous late downturn in the survival curve of patients with primary choroidal or ciliochoroidal melanoma treated by plaque radiotherapy and (2) the survival curves of patients with comparable choroidal and ciliochoroidal melanomas treated by plaque radiotherapy versus enucleation are quite similar over the first 15 years of post-treatment follow-up. These observations are in agreement with the results of our recently published study on an independent cohort of patients treated by cobalt-60 plaque radiotherapy or enucleation between 1976 and 1980 [5]. These observations are important because they contradict the published expectations of some advocates of enucleation [13, 14].

One aspect of our results, the finding that the survival rate among plaque-treated patients who experienced local tumor relapse was substantially higher than that among

plaque-treated patients who did not have local recurrence [9, 16], is particularly worrisome to many clinicians. The logical expectations, based on this finding, are (1) that the survival curve of plaque-treated patients should exhibit a marked delayed downturn compared to that of comparable patients treated by primary enucleation and (2) that the survival curves of comparable groups of patients treated by plaque versus enucleation should start to separate after about 7–10 years. The fact that these expectations are not fulfilled suggests that the implication of local relapse may differ from what is generally perceived. Our results suggest that local relapse following plaque therapy may be an indicator of more aggressive tumor histology at baseline and greater likelihood of occult hematogenous dissemination of tumor, which in turn translates into higher rates of subsequent clinical metastasis and tumor-related mortality. Because a comparable group of patients managed by primary enucleation would be expected to have a similar proportion of tumors that would have relapsed locally had they been treated by plaque radiotherapy, they would be expected to exhibit a similar time to development of clinical metastasis.

Our observations suggest that a large difference in melanoma-specific mortality rates between equivalent patients with a primary choroidal or ciliary body melanoma treated by plaque radiotherapy or by enucleation is unlikely. We anticipate that this result will be confirmed by the Collaborative Ocular Melanoma Study's randomized clinical trial of enucleation versus iodine-125 plaque radiotherapy for medium-size choroidal melanomas [15]. Yet, because enrollment in that trial did not begin in earnest until 1987, we do not anticipate that COMS will be able to report 15-year follow-up data on its first 100 or more enrolled patients until at least the year 2003 or 2004. Until then, the results of nonrandomized but statis-

tically adjusted retrospective survival studies such as this one should be regarded as the best available evidence about the relative effectiveness of plaque radiotherapy and enucleation with regard to melanoma-specific mortality.

Acknowledgements This work was supported in part by the Uveal Melanoma Study Fund of Wills Eye Hospital, Philadelphia, Pennsylvania, USA, the Vitreoretinal Research and Education Foundation, Philadelphia, Pennsylvania, USA, and Research to Prevent Blindness, Inc., New York, New York, USA.

References

1. Adams KS, Abramson DH, Ellsworth RM, Haik BG, Bedford M, Packer S, Seddon J, Albert D, Polivogianis L (1988) Cobalt plaque versus enucleation for uveal melanoma: comparison of survival rates. *Br J Ophthalmol* 72:494–497
2. Augsburger JJ, Gamel JW, Sardi VF, Greenberg RA, Shields JA, Brady LW (1986) Enucleation vs cobalt plaque radiotherapy for malignant melanomas of the choroid and ciliary body. *Arch Ophthalmol* 104:655–661
3. Augsburger JJ, Gamel JW, Shields JA (1989) Cobalt plaque radiotherapy versus enucleation for posterior uveal melanoma: comparison of survival by prognostic index groups. *Trans Am Ophthalmol Soc* 87:348–361
4. Augsburger JJ, Gamel JW, Lauritzen K, Brady LW (1990) Cobalt-60 plaque radiotherapy vs enucleation for posterior uveal melanoma. *Am J Ophthalmol* 109:585–592
5. Augsburger JJ, Corrêa Z, Freire J, Brady LW (1998) Long-term survival in posterior uveal melanoma after enucleation versus plaque radiotherapy. *Ophthalmology* 105:1670–1678
6. Feinstein AR (1975) The other side of 'statistical significance': alpha, beta, delta, and the calculation of sample size. *Clin Pharmacol Ther* 18:491–505
7. Giblin ME, Shields JA, Augsburger JJ, Brady LW (1989) Episcleral plaque radiotherapy for uveal melanoma. *Aust N Z J Ophthalmol* 17:153–156
8. Guthoff R, Frischmuth J, Jensen OA, Bjerrum K, Prause JU (1992) Das Aderhautmelanom. Eine retrospektive randomisierte Vergleichsstudie Ruthenium-Bestrahlung vs Enukleation. *Klin Monatsbl Augenheilkd* 200:257–261
9. Karlsson UL, Augsburger JJ, Shields JA, Markoe AM, Brady LW, Woodleigh R (1989) Recurrence of posterior uveal melanoma after ⁶⁰Co episcleral plaque therapy. *Ophthalmology* 96:382–388
10. Lee ET (1980) Life-table analysis. In: Lee ET (ed) *Statistical methods for survival data analysis*. Lifetime Learning, Belmont, Calif, pp 88–92
11. Lee ET (1980) Mantel and Haenszel's chi-square test. In: Lee ET (ed) *Statistical methods for survival data analysis*. Lifetime Learning, Belmont, Calif, pp 136–139
12. Lee ET (1980) Examining variables simultaneously – Cox's regression model for survival data. In: Lee ET (ed) *Statistical methods for survival data analysis*. Lifetime Learning, Belmont, Calif, pp 306–315
13. Manschot WA, van Strik R (1987) Is irradiation a justifiable treatment of choroidal melanoma? Analysis of published results. *Br J Ophthalmol* 71:348–352
14. Manschot WA, Lee WR, van Strik R (1996) Uveal melanoma: updated considerations on current management modalities. *Int Ophthalmol* 19:203–209
15. Straatsma BR, Fine SL, Earle JD, Hawkins BS, Diener-West M, McLaughlin JA, The Collaborative Ocular Melanoma Study Research Group (1988) Enucleation versus plaque irradiation for choroidal melanoma. *Ophthalmology* 95:1000–1004
16. Vrabec TR, Augsburger JJ, Gamel JW, Brady LW, Hernandez C, Woodleigh R (1991) Impact of local tumor relapse on patient survival after cobalt 60 plaque radiotherapy. *Ophthalmology* 98:984–988